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SECTION A

QUESTION 1

Ubuntu and Kali Linux are Linux based open-source Operating systems. With TechCorp focusing on web development and network security, this essay will focus on describing both Ubuntu and Kali Linux, outlining their advantages and disadvantages, and recommending the best OS between the two to TechCorp with regard to their focus areas.

Ubuntu is based on the Debian Linux open supply running system. Since ubuntu is open source, it is free of cost, and everyone can make contributions to its improvements. It is absolutely unfastened to download and use. Ubuntu has a couple of versions, catering to numerous desired uses. The standard version is perfect to use for non-public systems, whilst Ubuntu Server is best for web sites or applications. Along with its user-friendliness, Ubuntu offers wide hardware compatibility, running with good range of peripheral devices and computer systems. Since Ubuntu is open-source, it allows for a great degree of customization. Users are able to install new topics and desktop environments and customize it however they want to. With the regular security updates and robust permission system, Ubuntu is also known for its strong security. Although Ubuntu consists of numerous advantages, it also has disadvantages. These include software compatibility, limited hardware support, learning curve and software updates. Software compatibility issues refer to some software not being compatible with Ubuntu, especially applications specifically made for Windows. Although Ubuntu supports a wide variety of hardware, in other cases support for specific hardware drivers is limited. With users accustomed to other operating systems, there may be a learning curve when it comes to switching to Ubuntu.

Ubuntu is used in numerous, including in, non-public computer systems, servers, internet improvement, open supply projects, education, net of factors (IoT), gaming and even in artificial intelligence. Non-public computer systems are a classic Ubuntu use case. The Ubuntu Server is a powerful server known for powering websites, applications, and cloud infrastructure. It provides stability, protection functions and sizeable software program environment making it perfect for organizations. Due to its loose nature, strong network support, and compatibility with diverse open-supply, a lot of open-supply tasks are dependent on Ubuntu for improvement and deployment. The open-source nature and powerful computational abilities make Ubuntu a suitable platform for AI and ML tasks.

Kali Linux, which like Ubuntu is a Debian based Linux distribution, is maintained by Offensive Security. Developers Mati Aharoni and Devon Kearns specially designed the OS for cybersecurity and analysis professionals. Kali Linux has four key features. These are penetration testing tools, forensic tools, wireless network tools, security and privacy features. The main tools of Kali Linux are its penetration tools. These tools help simulate real-world cyberattacks, which is quite essential if you are a cybersecurity professional. Kali Linux also has numerous forensic tools that help analyse digital evidence and recover lost data. Kali Linux provides numerous tools for wireless penetration testing. Lastly, the operating system is designed with built-in security mechanisms, ensuring a solid and robust environment.

A few advantages of Kali Linux include more than 600 tools for penetration testing, multi-language support, open development tree, and advanced security features. Kali Linux has more than 600 penetration-testing tools which are made to make it easier to check for the safety of a network. With its comprehensive multilingual support, Kali Linux allows more people to work in their native language. It also makes many products even better by the availability of free, open-source software. Kali Linux also has many state-of-the-art security features not found in other distributions. Although Kali Linux consists of numerous advantages, it too still has its disadvantages. These include that it is not intended for beginners, lacks documentation, is a bit slower than other distributions, and isn't suitable for all

hardware. Kali Linux is not meant to be used by beginners; it was designed for experienced users. It doesn't have a graphical user interface, so all tasks are done from the command line, and this may be too difficult for some people. It also lacks documentation which can make it difficult to learn how to use the tools. Because it uses many resources for penetration testing, Kali Linux is slower than other distributions. Lastly, Kali Linux isn't suitable for all hardware. Mobile devices can't run Kali Linux.

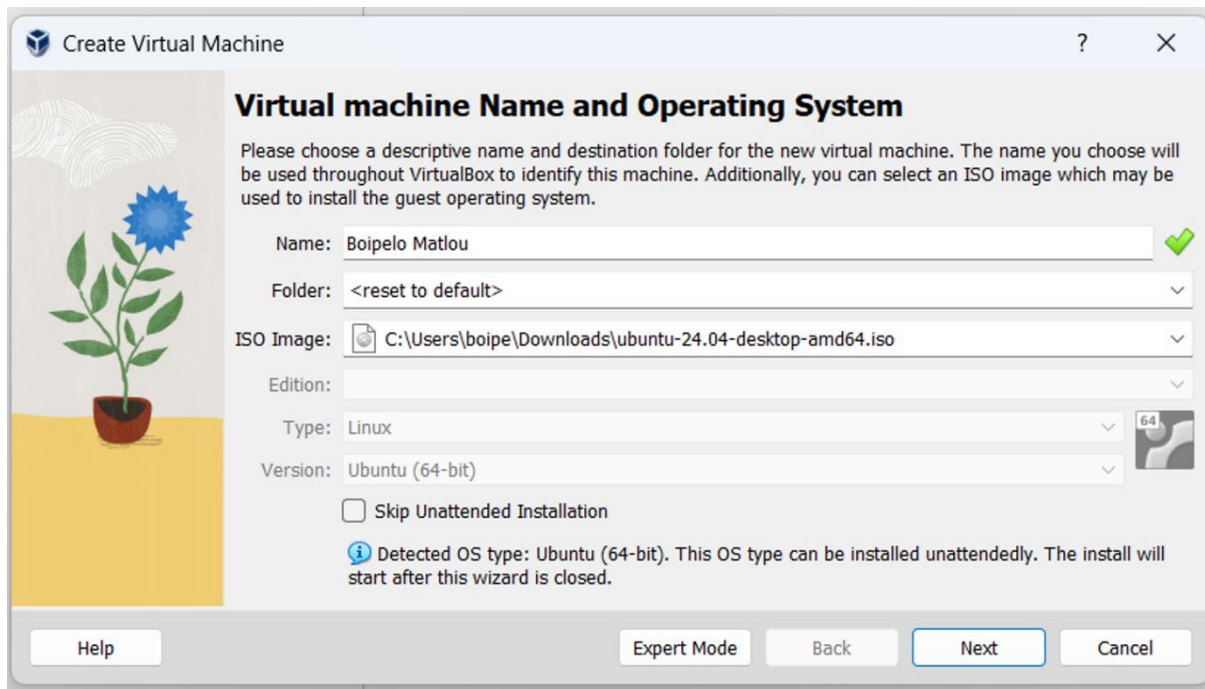
In conclusion, with regard to the fields TechCorp focuses on, I recommend Kali Linux over Ubuntu. Although Ubuntu is a phenomenal Linux distribution, Kali Linux and Ubuntu serve different purpose. Ubuntu is more suitable for beginner users who want to learn Linux and experienced programmers. TechCorp require a distribution that is most suitable for web development, network security and penetration testing. Kali Linux has over 600 tools for penetration testing and has state-of-the-art security features like no other distribution. Kali Linux was designed for network analysts and cybersecurity professionals, to better simulate real-world cyberattacks, and perform penetration testing.

QUESTION 2

2.1 VM Setup

Create a VM using a virtualization tool of your choice

Name the VM you Student Name



Create Virtual Machine

Virtual machine Name and Operating System

Please choose a descriptive name and destination folder for the new virtual machine. The name you choose will be used throughout VirtualBox to identify this machine. Additionally, you can select an ISO image which may be used to install the guest operating system.

Name: Boipelo Matlou ✓

Folder: <reset to default>

ISO Image: C:\Users\boipe\Downloads\ubuntu-24.04-desktop-amd64.iso

Edition:

Type: Linux

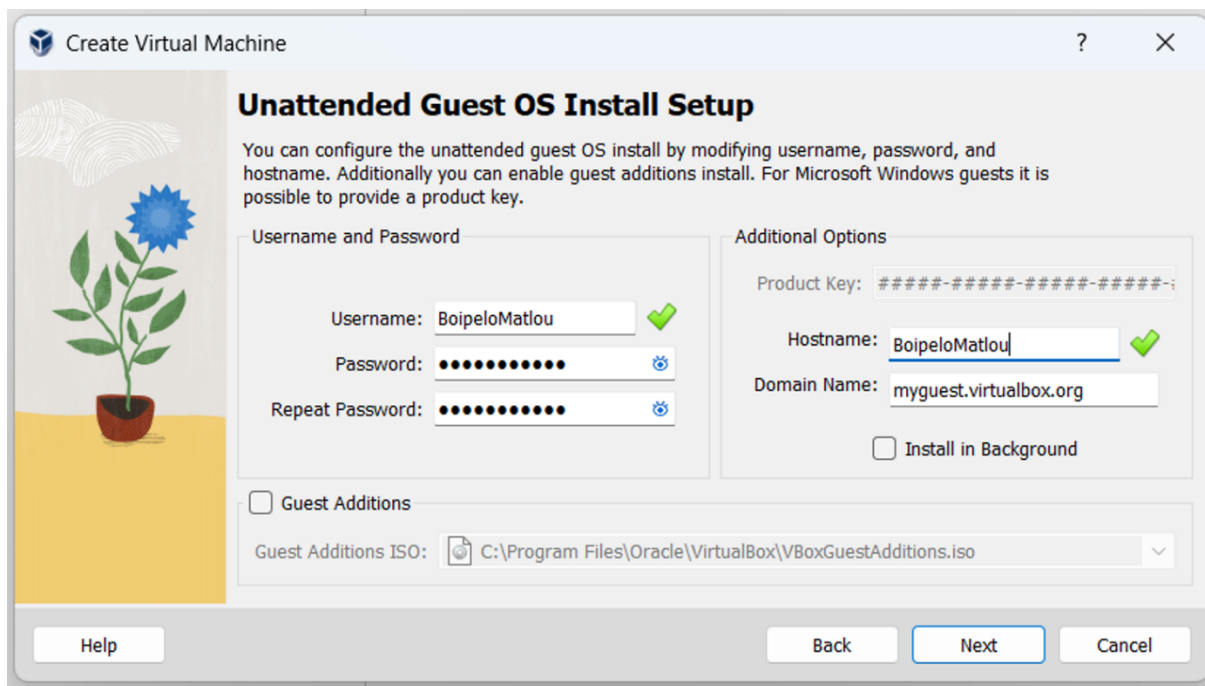
Version: Ubuntu (64-bit)

☐ Skip Unattended Installation

Detected OS type: Ubuntu (64-bit). This OS type can be installed unattendedly. The install will start after this wizard is closed.

Help Expert Mode Back Next Cancel

Install Ubuntu Linux as the guest operating system



Create Virtual Machine

Unattended Guest OS Install Setup

You can configure the unattended guest OS install by modifying username, password, and hostname. Additionally you can enable guest additions install. For Microsoft Windows guests it is possible to provide a product key.

Username and Password

Username: BoipeloMatlou ✓

Password: ●●●●●●●●

Repeat Password: ●●●●●●●●

☐ Guest Additions

Guest Additions ISO: C:\Program Files\Oracle\VirtualBox\VBBoxGuestAdditions.iso

Additional Options

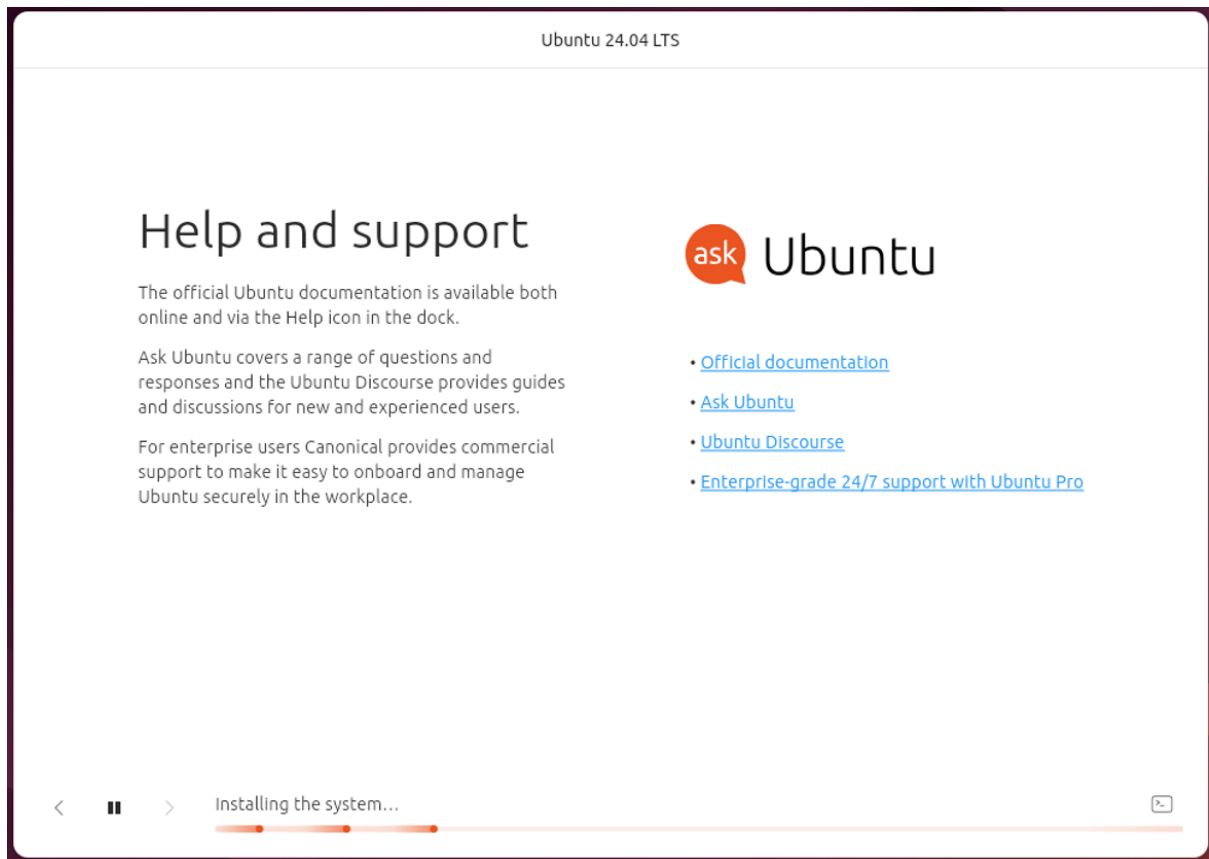
Product Key: #####-#####-#####-#####-;

Hostname: BoipeloMatlou ✓

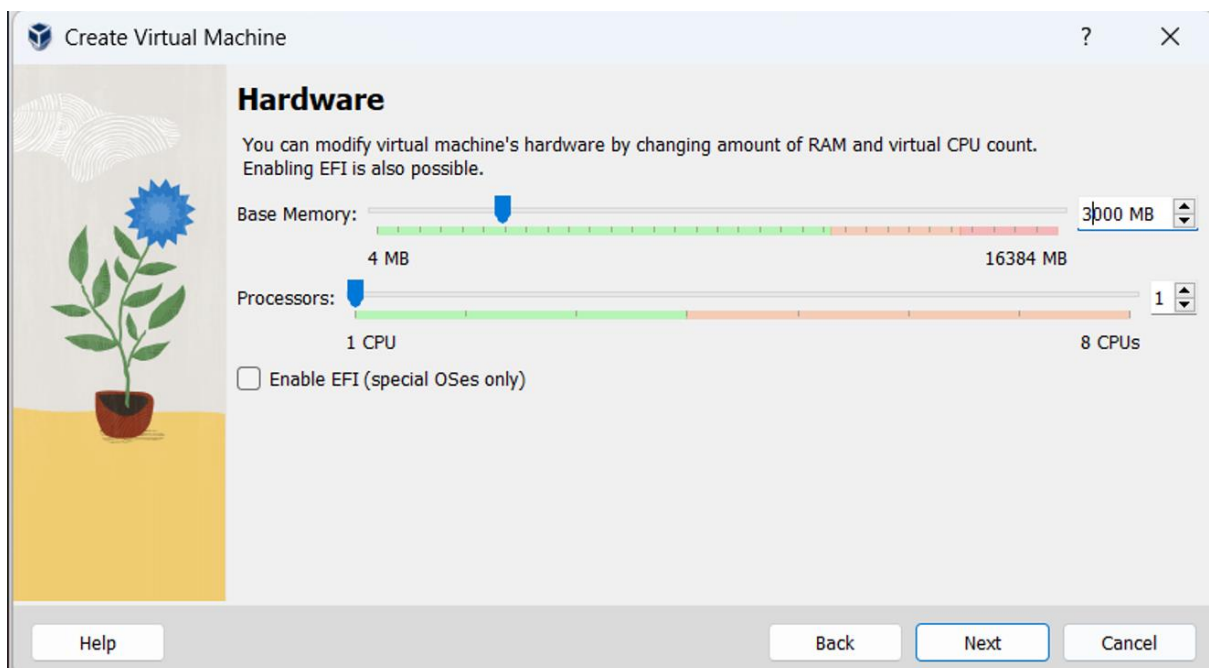
Domain Name: myguest.virtualbox.org

☐ Install in Background

Help Back Next Cancel



Configure the VM with at least 2 GB of RAM, 20 GB of disk space, and network connectivity



Create Virtual Machine

Virtual Hard disk

If you wish you can add a virtual hard disk to the new machine. You can either create a new hard disk file or select an existing one. Alternatively you can create a virtual machine without a virtual hard disk.

☒ Create a Virtual Hard Disk Now

Disk Size:

4,00 MB 2,00 TB

☐ Pre-allocate Full Size

☐ Use an Existing Virtual Hard Disk File

☐ Do Not Add a Virtual Hard Disk

Help Back Next Cancel

Create Virtual Machine

Summary

The following table summarizes the configuration you have chosen for the new virtual machine. When you are happy with the configuration press Finish to create the virtual machine. Alternatively you can go back and modify the configuration.

Machine Name and OS Type	
Machine Name	Boipelo Matlou
Machine Folder	C:/Users/boipe/.VirtualBox/o/Boipelo Matlou
ISO Image	C:/Users/boipe/Downloads/ubuntu-24.04-desktop-amd64.iso
Guest OS Type	Ubuntu (64-bit)
Skip Unattended Install	false
Unattended Install	
Username	BoipeloMatlou
Product Key	false
Hostname/Domain Name	BoipeloMatlou.myguest.virtualbox.org
Install in Background	false
Install Guest Additions	false
Hardware	
Base Memory	3000
Processor(s)	1
EFI Enable	false
Disk	
Disk Size	25,00 GB
Pre-allocate Full Size	false

Help Back Finish Cancel

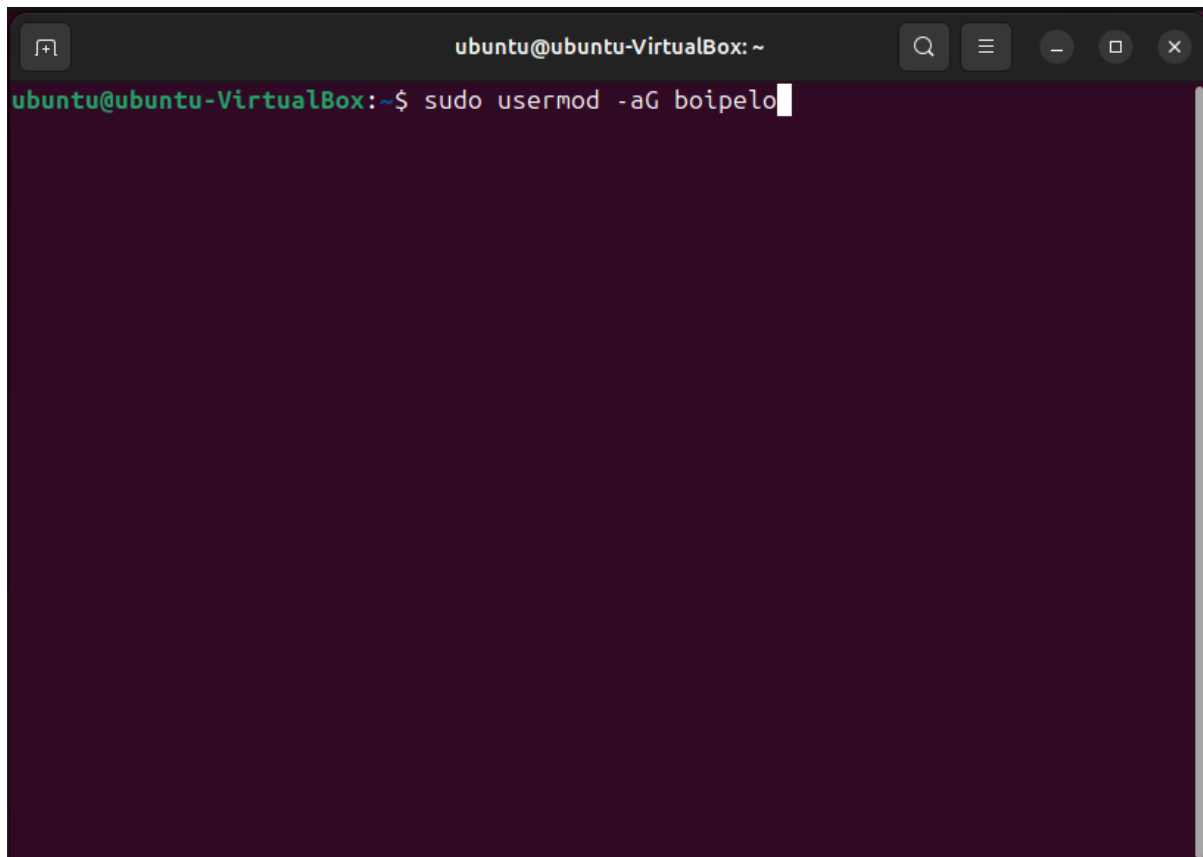
Summary

2.2 User and Permissions

Create new user with your name as the username

```
ubuntu@ubuntu-VirtualBox:~$ sudo adduser boipelo
info: Adding user `boipelo' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `boipelo' (1001) ...
info: Adding new user `boipelo' (1001) with group `boipelo (1001)' ...
info: Creating home directory `/home/boipelo' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for boipelo
Enter the new value, or press ENTER for the default
    Full Name []: Boipelo Matlou
    Room Number []: 434
    Work Phone []:
    Home Phone []:
    Other []: y
Is the information correct? [Y/n] y
info: Adding new user `boipelo' to supplemental / extra groups `users' ...
info: Adding user `boipelo' to group `users' ...
ubuntu@ubuntu-VirtualBox:~$
```

Grant them sudo privileges



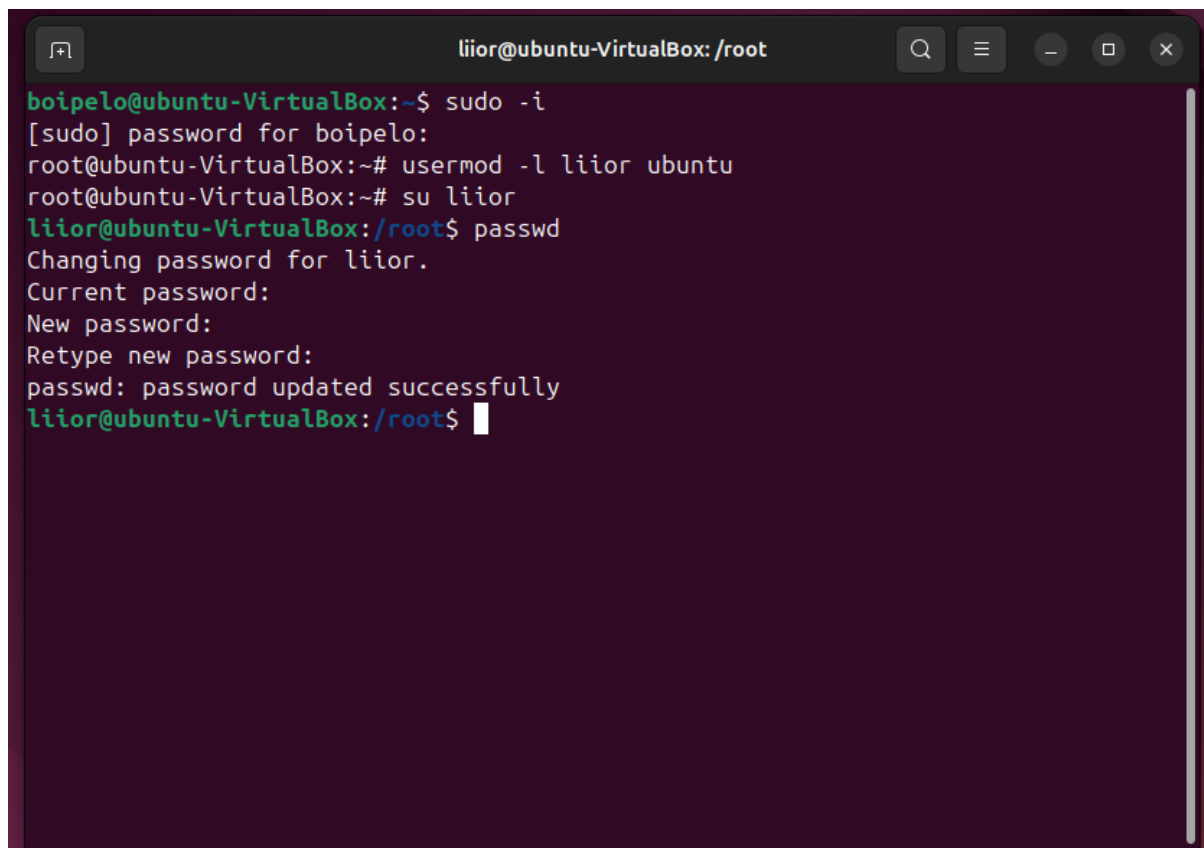
A terminal window titled 'ubuntu@ubuntu-VirtualBox: ~' with standard window controls (search, menu, zoom, close). The terminal shows the command 'sudo usermod -aG boipelo' being entered at the prompt. The command is partially executed, with a cursor at the end of the line.

```
ubuntu@ubuntu-VirtualBox:~$ sudo usermod -aG boipelo
```

```
ubuntu@ubuntu-VirtualBox: ~  
GNU nano 7.2 /etc/sudoers.tmp  
# Host alias specification  
  
# User alias specification  
  
# Cmnd alias specification  
  
# User privilege specification  
root    ALL=(ALL:ALL) ALL  
  
# Members of the admin group may gain root privileges  
%admin   ALL=(ALL) ALL  
  
# Allow members of group sudo to execute any command  
%sudo    ALL=(ALL:ALL) ALL  
  
# See sudoers(5) for more information on "@include" directives:  
  
@includedir /etc/sudoers.d  
boipelo ALL=(ALL) ALL  
  
^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute  ^C Location  
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify  ^/ Go To Line
```

```
ubuntu@ubuntu-VirtualBox:~$ su boipelo  
Password:  
boipelo@ubuntu-VirtualBox:/home/ubuntu$ sudo adduser bakwe  
[sudo] password for boipelo:  
info: Adding user `bakwe' ...  
info: Selecting UID/GID from range 1000 to 59999 ...  
info: Adding new group `bakwe' (1002) ...  
info: Adding new user `bakwe' (1002) with group `bakwe (1002)' ...  
info: Creating home directory `/home/bakwe' ...  
info: Copying files from `/etc/skel' ...  
New password: 
```

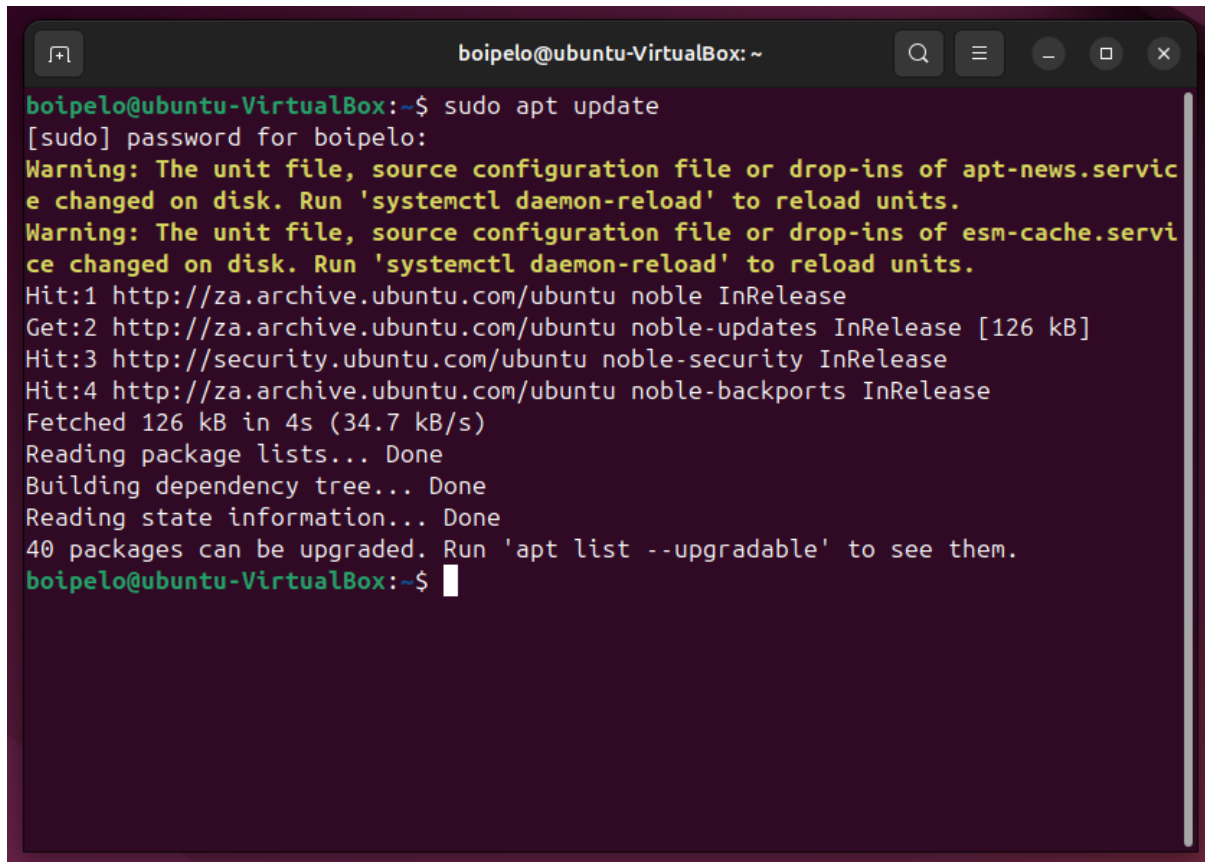
Modify default user (ubuntu) to have a custom username and set a secure password



```
liior@ubuntu-VirtualBox: /root
boipelo@ubuntu-VirtualBox:~$ sudo -i
[sudo] password for boipelo:
root@ubuntu-VirtualBox:~# usermod -l liior ubuntu
root@ubuntu-VirtualBox:~# su liior
liior@ubuntu-VirtualBox:/root$ passwd
Changing password for liior.
Current password:
New password:
Retype new password:
passwd: password updated successfully
liior@ubuntu-VirtualBox:/root$
```

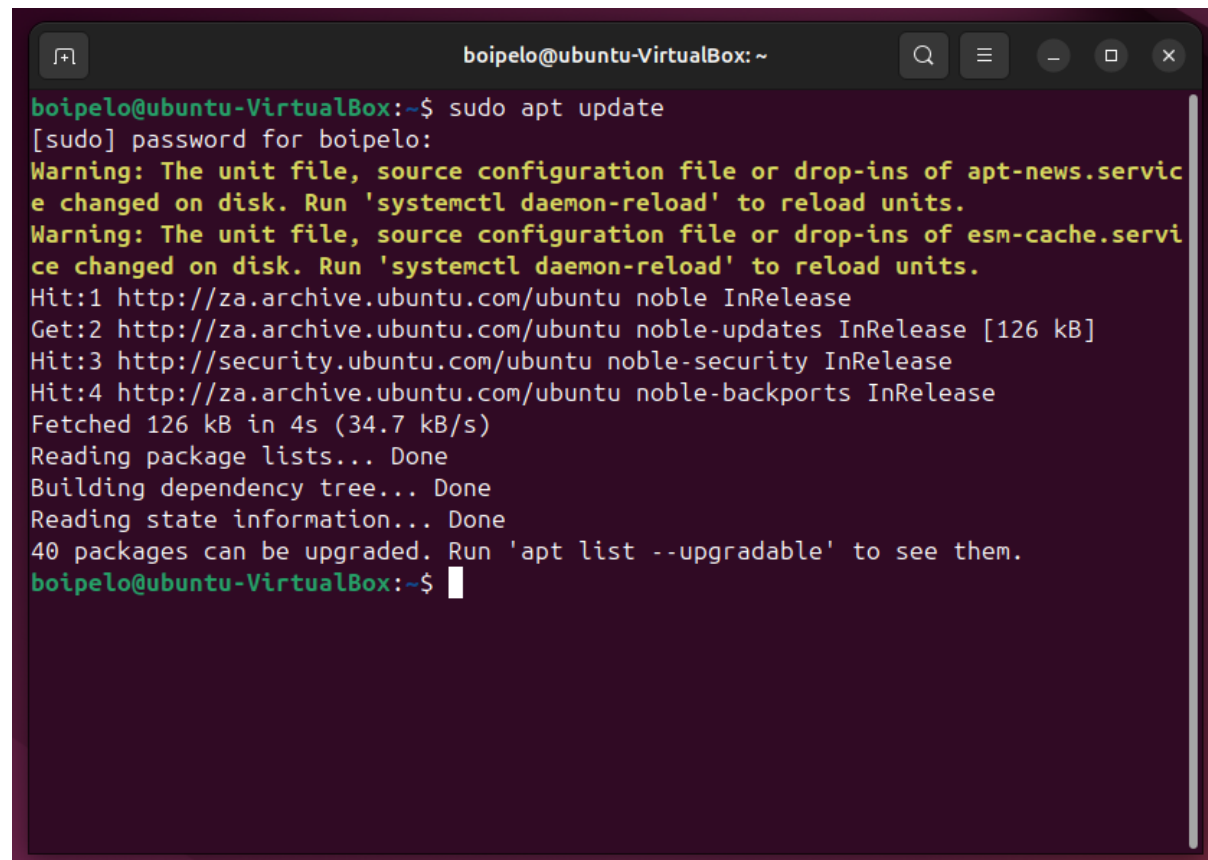
2.3 Terminal Tasks

Update system packages

A terminal window titled 'boipelo@ubuntu-VirtualBox: ~' with standard window controls. The terminal shows the execution of 'sudo apt update'. It prompts for a password, displays two warnings about unit files for 'apt-news.service' and 'esm-cache.service', and then shows the progress of updating package lists from various mirrors. It reports that 40 packages can be upgraded and suggests running 'apt list --upgradable' to see them. The prompt returns to the user.

```
boipelo@ubuntu-VirtualBox:~$ sudo apt update
[sudo] password for boipelo:
Warning: The unit file, source configuration file or drop-ins of apt-news.service
changed on disk. Run 'systemctl daemon-reload' to reload units.
Warning: The unit file, source configuration file or drop-ins of esm-cache.servi
ce changed on disk. Run 'systemctl daemon-reload' to reload units.
Hit:1 http://za.archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://za.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Hit:3 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:4 http://za.archive.ubuntu.com/ubuntu noble-backports InRelease
Fetched 126 kB in 4s (34.7 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
40 packages can be upgraded. Run 'apt list --upgradable' to see them.
boipelo@ubuntu-VirtualBox:~$
```

Install chrome using the terminal

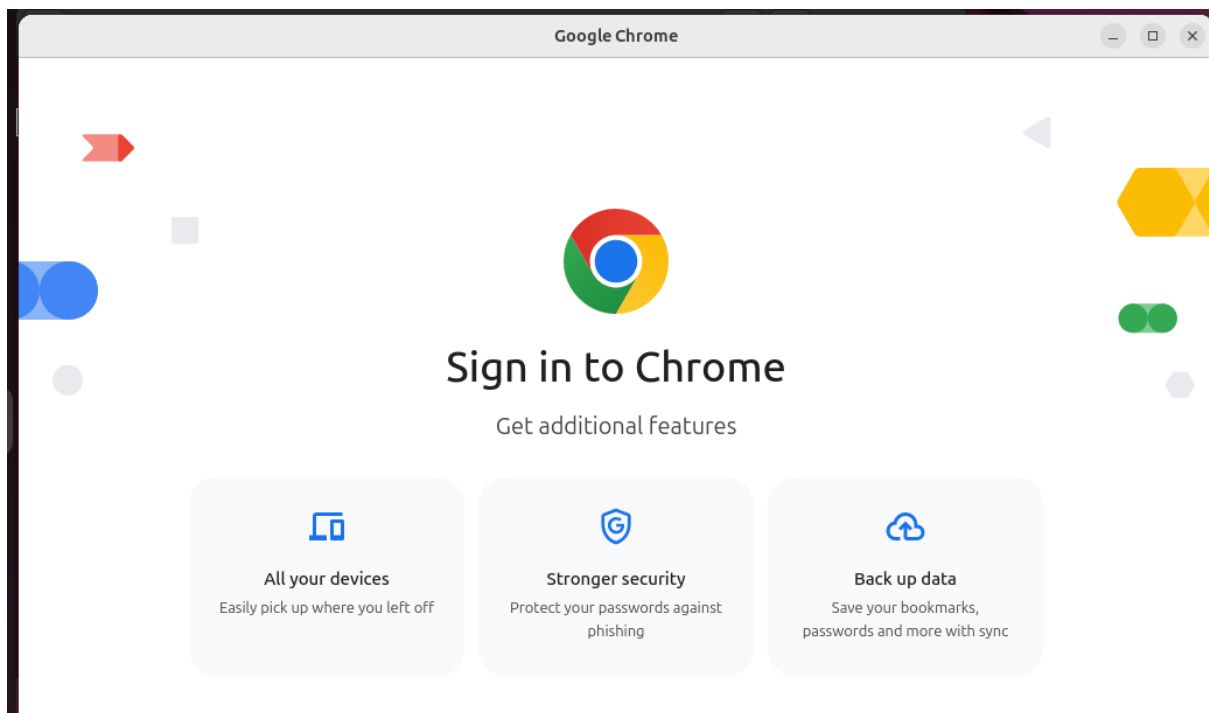
A terminal window titled 'boipelo@ubuntu-VirtualBox: ~' with standard window controls. The terminal shows the execution of 'sudo apt update'. It prompts for a password, displays two warnings about unit files, and shows the progress of updating package lists from various mirrors. It reports that 40 packages can be upgraded.

```
boipelo@ubuntu-VirtualBox:~$ sudo apt update
[sudo] password for boipelo:
Warning: The unit file, source configuration file or drop-ins of apt-news.servic
e changed on disk. Run 'systemctl daemon-reload' to reload units.
Warning: The unit file, source configuration file or drop-ins of esm-cache.servi
ce changed on disk. Run 'systemctl daemon-reload' to reload units.
Hit:1 http://za.archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://za.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Hit:3 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:4 http://za.archive.ubuntu.com/ubuntu noble-backports InRelease
Fetched 126 kB in 4s (34.7 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
40 packages can be upgraded. Run 'apt list --upgradable' to see them.
boipelo@ubuntu-VirtualBox:~$
```

```
boipelo@ubuntu-VirtualBox: ~  
boipelo@ubuntu-VirtualBox:~$ sudo apt upgrade  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
Calculating upgrade... Done  
The following packages will be upgraded:  
  apport apport-core-dump-handler apport-gtk cloud-init dhcpcd-base  
  gnome-shell-extension-appindicator initramfs-tools initramfs-tools-bin  
  initramfs-tools-core libnautilus-extension4 libnss-systemd libpam-modules  
  libpam-modules-bin libpam-runtime libpam-systemd libpam0g libsystemd-shared  
  libsystemd0 libudev1 nautilus nautilus-data python3-apport  
  python3-distupgrade python3-problem-report snapd systemd systemd-dev  
  systemd-oom systemd-resolved systemd-sysv systemd-timesyncd tzdata  
  ubuntu-pro-client ubuntu-pro-client-l10n ubuntu-release-upgrader-core  
  ubuntu-release-upgrader-gtk udev vim-common vim-tiny xxd  
40 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.  
Need to get 41.1 MB of archives.  
After this operation, 2,995 kB of additional disk space will be used.  
Do you want to continue? [Y/n] y  
Get:1 http://za.archive.ubuntu.com/ubuntu noble-updates/main amd64 libpam0g amd6  
4 1.5.3-5ubuntu5.1 [67.7 kB]  
Get:2 http://za.archive.ubuntu.com/ubuntu noble-updates/main amd64 systemd-times  
yncd amd64 255.4-1ubuntu8.1 [35.3 kB]  
Get:3 http://za.archive.ubuntu.com/ubuntu noble-updates/main amd64 systemd-resol
```

```
boipelo@ubuntu-VirtualBox:~$ wget https://dl.google.com/linux/direct/google-chro  
me-stable_current_amd64.deb  
--2024-06-21 09:12:49-- https://dl.google.com/linux/direct/google-chrome-stable  
_current_amd64.deb  
Resolving dl.google.com (dl.google.com)... 142.251.47.142, 2c0f:fb50:4002:808::2  
00e  
Connecting to dl.google.com (dl.google.com)|142.251.47.142|:443... connected.  
HTTP request sent, awaiting response... 200 OK  
Length: 108805872 (104M) [application/x-debian-package]  
Saving to: 'google-chrome-stable_current_amd64.deb'  
  
google-chrome-stabl 100%[=====>] 103.76M 1.06MB/s in 2m 3s  
  
2024-06-21 09:14:52 (867 KB/s) - 'google-chrome-stable_current_amd64.deb' saved  
[108805872/108805872]  
  
boipelo@ubuntu-VirtualBox:~$
```

```
boipelo@ubuntu-VirtualBox:~$ sudo dpkg -i google-chrome-stable_current_amd64.deb
Selecting previously unselected package google-chrome-stable.
(Reading database ... 147881 files and directories currently installed.)
Preparing to unpack google-chrome-stable_current_amd64.deb ...
Unpacking google-chrome-stable (126.0.6478.114-1) ...
Setting up google-chrome-stable (126.0.6478.114-1) ...
update-alternatives: using /usr/bin/google-chrome-stable to provide /usr/bin/x-w
ww-browser (x-www-browser) in auto mode
update-alternatives: using /usr/bin/google-chrome-stable to provide /usr/bin/gno
me-www-browser (gnome-www-browser) in auto mode
update-alternatives: using /usr/bin/google-chrome-stable to provide /usr/bin/goo
gle-chrome (google-chrome) in auto mode
Processing triggers for gnome-menus (3.36.0-1ubuntu3) ...
Processing triggers for desktop-file-utils (0.27-2build1) ...
Processing triggers for man-db (2.12.0-4build2) ...
boipelo@ubuntu-VirtualBox:~$ sudo apt-get install -f
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
boipelo@ubuntu-VirtualBox:~$ google-chrome
```



Configure the system to display hidden files and folders in the file manager

```
boipelo@ubuntu-VirtualBox:~$ ls -a
.          .cache      google-chrome-stable_current_amd64.deb  .profile
..         .config     .local                                   Public
.bash_history Desktop    Music                                   snap
.bash_logout Documents Pictures                               Templates
.bashrc     Downloads .pki                                    Videos
boipelo@ubuntu-VirtualBox:~$
```

Change hostname to "Student Name and surname"

```
boipelo@ubuntu-VirtualBox:~$ hostnamectl set-hostname BoipeloMatlou
boipelo@ubuntu-VirtualBox:~$ hostname
BoipeloMatlou
boipelo@ubuntu-VirtualBox:~$
```

Create a text file in the terminal and store in work documents directory

```
boipelo@ubuntu-VirtualBox:~$ cd Documents
boipelo@ubuntu-VirtualBox:~/Documents$ dir
ITLSA  Work
boipelo@ubuntu-VirtualBox:~/Documents$ cd Work
boipelo@ubuntu-VirtualBox:~/Documents/Work$ touch newtextfile
boipelo@ubuntu-VirtualBox:~/Documents/Work$ ls -a
.  ..  newtextfile
boipelo@ubuntu-VirtualBox:~/Documents/Work$
```

View text file with terminal without opening the entire file

```
boipelo@ubuntu-VirtualBox:~/Documents/Work$ more newtextfile
```

2.4 Setup a basic firewall to restrict incoming network traffic

```
boipelo@ubuntu-VirtualBox: ~/Documents/Work
boipelo@ubuntu-VirtualBox:~/Documents/Work$ sudo apt install iptables-persistent
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  netfilter-persistent
The following packages will be REMOVED:
  ufw
The following NEW packages will be installed:
  iptables-persistent netfilter-persistent
0 upgraded, 2 newly installed, 1 to remove and 0 not upgraded.
Need to get 14.3 kB of archives.
After this operation, 780 kB disk space will be freed.
Do you want to continue? [Y/n] y
Get:1 http://za.archive.ubuntu.com/ubuntu noble/universe amd64 netfilter-persistent all 1.0.20 [7,402 B]
Get:2 http://za.archive.ubuntu.com/ubuntu noble/universe amd64 iptables-persistent all 1.0.20 [6,946 B]
Fetched 14.3 kB in 2s (9,044 B/s)
Preconfiguring packages ...
(Reading database ... 147995 files and directories currently installed.)
Removing ufw (0.36.2-6) ...
Skip stopping firewall: ufw (not enabled)
Selecting previously unselected package netfilter-persistent.
```

```
boipelo@ubuntu-VirtualBox: ~/Documents/Work
GNU nano 7.2 /etc/iptables/rules.v4 *
sudo iptables -F
sudo iptables -X

# Default policies to drop
sudo iptables -P INPUT DROP
sudo iptables -P FORWARD DROP
sudo iptables -P OUTPUT DROP

# Localhost traffic allowed
sudo iptables -A INPUT -i lo -j ACCEPT

#Established and related incoming traffi allowed
sudo iptables -A INPUT -m conntrack --ctstate ESTABLISHED,RELATED -j ACCEPT

# Save rules
sudo sh -c "iptables-save > /etc/iptables/rules.v4"

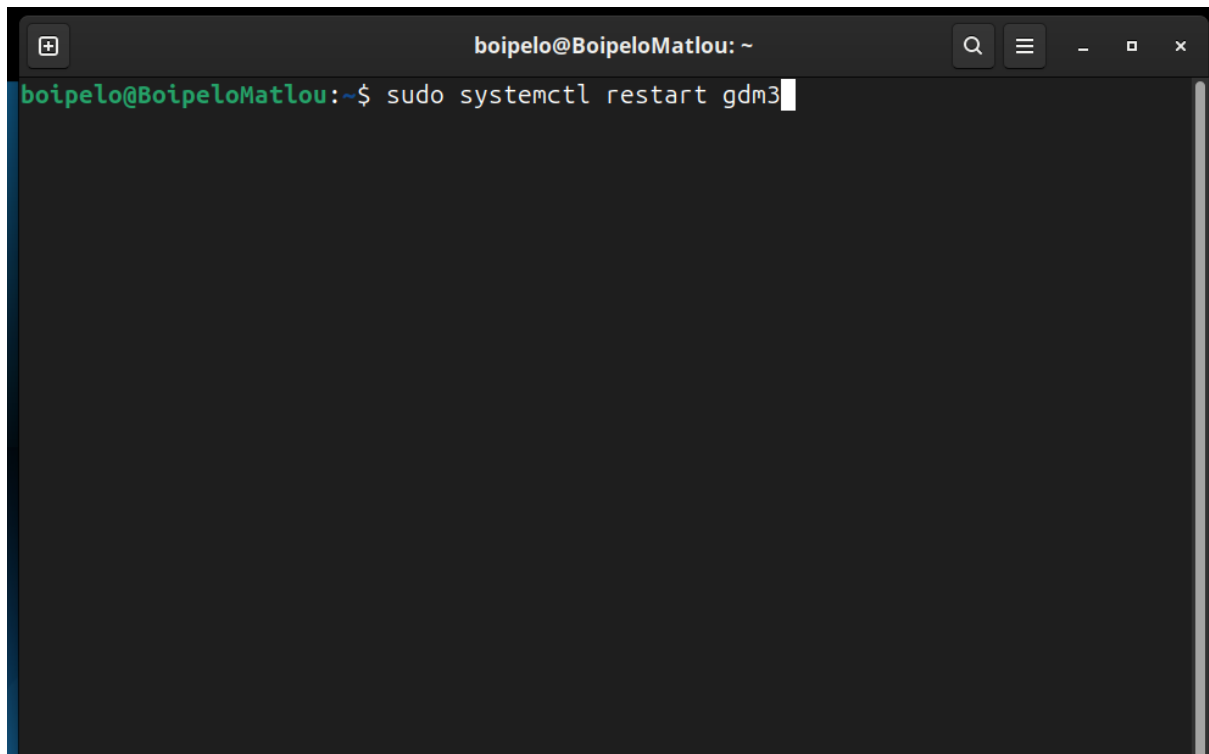
File Name to Write: /etc/iptables/rules.v4
^G Help          M-D DOS Format   M-A Append       M-B Backup File
^C Cancel        M-M Mac Format   M-P Prepend      ^T Browse
```

QUESTION 3

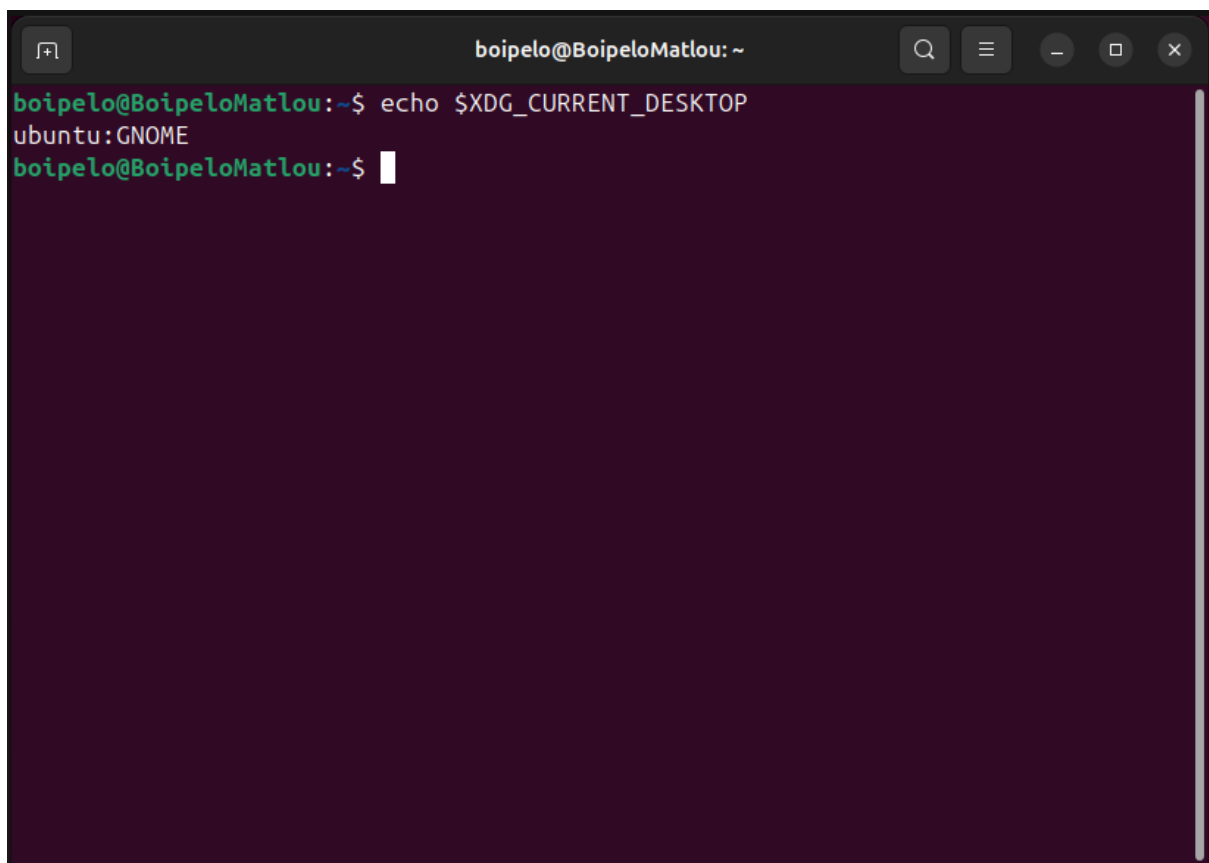
3.1) Installing Ubuntu GNOME desktop environment

```
boipelo@BoipeloMatlou:~$ sudo apt update
[sudo] password for boipelo:
Hit:1 http://za.archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://za.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Hit:3 http://security.ubuntu.com/ubuntu noble-security InRelease
Ign:4 http://archive.ubuntu.com jammy InRelease
Hit:5 http://za.archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:6 https://dl.google.com/linux/chrome/deb stable InRelease
Err:7 http://archive.ubuntu.com jammy Release
  404 Not Found [IP: 91.189.91.83 80]
Reading package lists... Done
E: The repository 'http://archive.ubuntu.com jammy Release' does not have a Release file.
N: Updating from such a repository can't be done securely, and is therefore disabled by default.
N: See apt-secure(8) manpage for repository creation and user configuration details.
boipelo@BoipeloMatlou:~$
```

```
boipelo@BoipeloMatlou:~$ sudo apt install ubuntu-gnome-desktop
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  bzip2 deja-dup duplicity file-roller fonts-cantarell fonts-opensymbol
  gir1.2-dbusmenu-glib-0.4 gir1.2-dee-1.0 gir1.2-gst-plugins-base-1.0
  gir1.2-gudev-1.0 gir1.2-rb-3.0 gir1.2-totem-1.0 gir1.2-totemlparser-1.0
  gir1.2-udisks-2.0 gir1.2-unity-7.0 gnome-calendar gnome-session
  gnome-snapshot grilo-plugins-0.3-base gstreamer1.0-gtk3
  gstreamer1.0-libcamera libabw-0.1-1 libavahi-ui-gtk3-0
  libboost-iostreams1.83.0 libboost-locale1.83.0 libboost-thread1.83.0
  libburn4t64 libcamera0.2 libcdr-0.1-1 libclucene-contribs1t64
  libclucene-core1t64 libcolamd3 libdmapsharing-4.0-3t64 libe-book-0.1-1
  libedataserverui4-1.0-0t64 libeot0 libepubgen-0.1-1 libetonyek-0.1-1
  libevent-2.1-7t64 libexttextcat-2.0-0 libexttextcat-data libfreehand-0.1-1
  libfreerdp-client3-3 libgom-1.0-0t64 libgpgmepp6t64 libgpod-common
  libgpod4t64 libgrilo-0.3-0 libisoburn1t64 libisofs6t64 liblangtag-common
  liblangtag1 liblirc-client0t64 liblttng-ust-common1t64 liblttng-ust-ctl5t64
  liblttng-ust1t64 libmhash2 libminiupnpc17 libmsspub-0.1-1 libmwaw-0.3-3
  libmythes-1.2-0 libnatpmp1t64 libodfgen-0.1-1 liborcus-0.18-0
  liborcus-parser-0.18-0 libpagemaker-0.0-0 libportal-gtk3-1 libraptor2-0
```



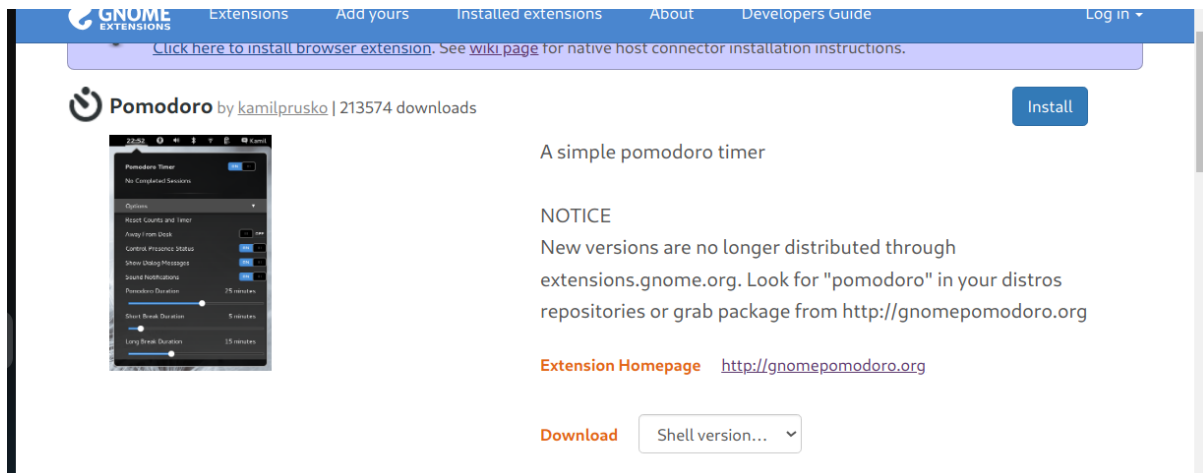
```
boipelo@BoipeloMatlou: ~  
boipelo@BoipeloMatlou:~$ sudo systemctl restart gdm3
```



```
boipelo@BoipeloMatlou: ~  
boipelo@BoipeloMatlou:~$ echo $XDG_CURRENT_DESKTOP  
ubuntu:GNOME  
boipelo@BoipeloMatlou:~$
```

3.2)

Pomodoro



Install the shell version of Pomodoro extension on the GNOME extensions site.

Opening Pomodoro extension on the terminal

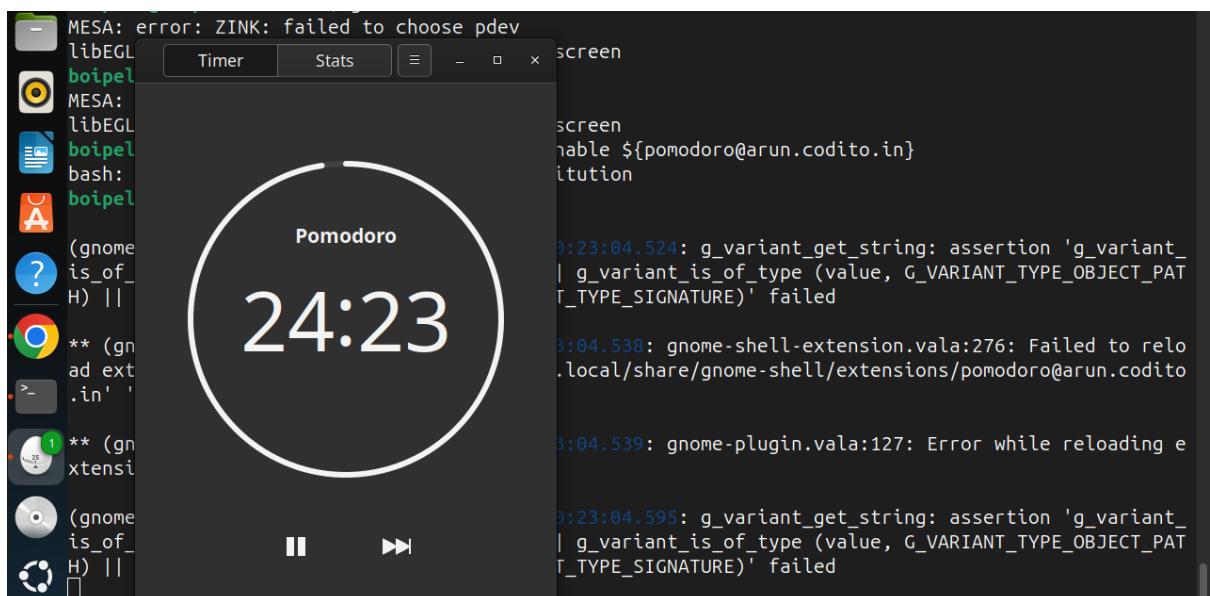
```
boipelo@BoipeloMatlou:~$ gnome-pomodoro

(gnome-pomodoro:14237): GLib-CRITICAL **: 20:23:04.524: g_variant_get_string: assertion 'g_variant_is_of_type (value, G_VARIANT_TYPE_STRING) || g_variant_is_of_type (value, G_VARIANT_TYPE_OBJECT_PATH) || g_variant_is_of_type (value, G_VARIANT_TYPE_SIGNATURE)' failed

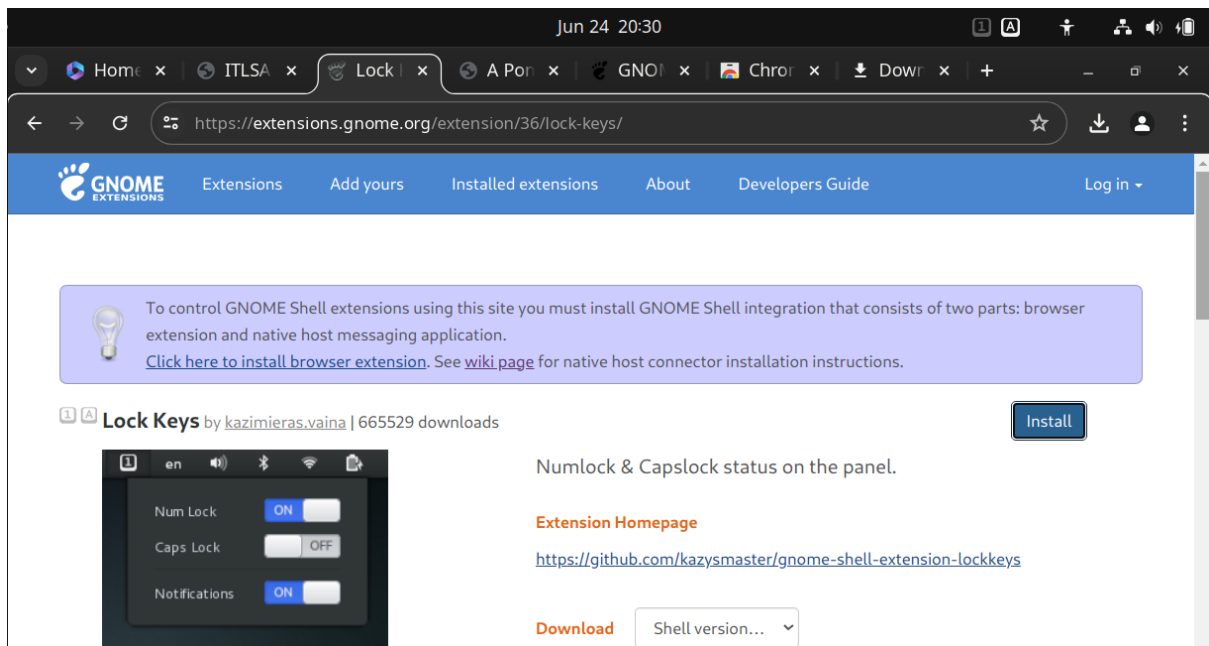
** (gnome-pomodoro:14237): WARNING **: 20:23:04.538: gnome-shell-extension.vala:276: Failed to reload extension: , out-of-date '/home/boipelo/.local/share/gnome-shell/extensions/pomodoro@arun.codito.in' '(null)'

** (gnome-pomodoro:14237): WARNING **: 20:23:04.539: gnome-plugin.vala:127: Error while reloading extension:
```

Functional Pomodoro

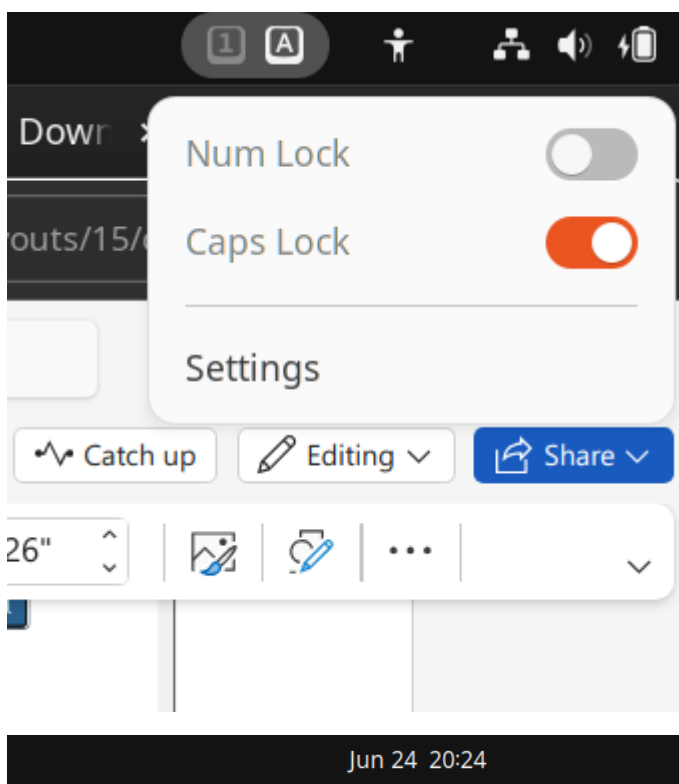


Lock Keys



Install the shell version of Lock Keys extension on the GNOME extensions site.

Functional Lock Keys



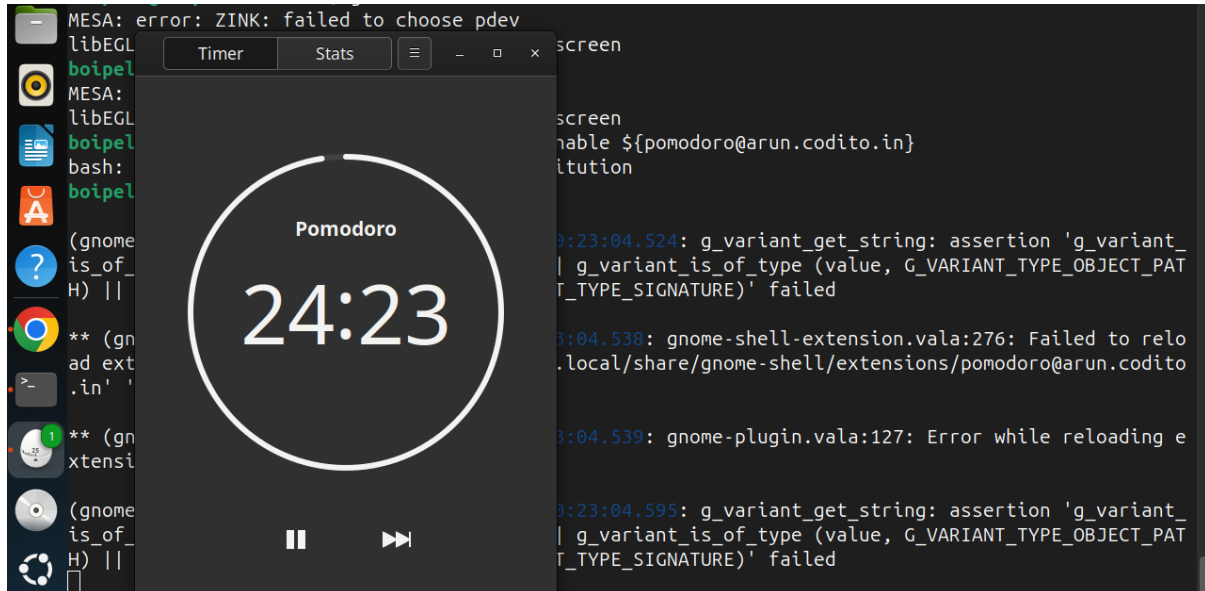
Pomodoro

The Pomodoro extension helps increase user productivity. It uses the Pomodoro technique that breaks work into intervals. A timer displays a countdown for work and break intervals. Notifications let the user know when a work or break interval begins. This extension allows users to set their own length for work and break intervals. As TechCorp focuses on web development, this extension could be an essential in increasing productivity for their web developers, breaking coding sessions up into intervals.

Installing Pomodoro:

- Go to [GNOME Shell Extensions](#)
- Search for Pomodoro in the search bar
- Install the Shell Version of the Pomodoro extension

Screenshot of functional Pomodoro extension:



Lock Keys

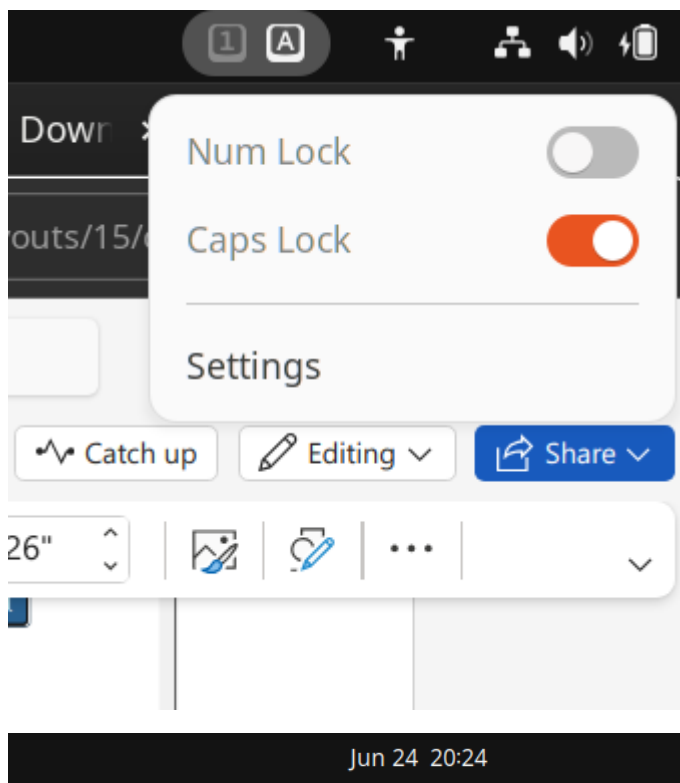
The Lock Keys extension displays the status of your Caps Lock, Num Lock, and Scroll Lock keys. This is an extension TechCorp web developers would benefit from. With the Lock Keys being on your screen, a mistake of entering a password or code incorrectly due to the Caps Lock being on, becomes very less common. Notifications

are optional on this extension. A user can configure the Lock Key to notify them whenever a key's status changes.

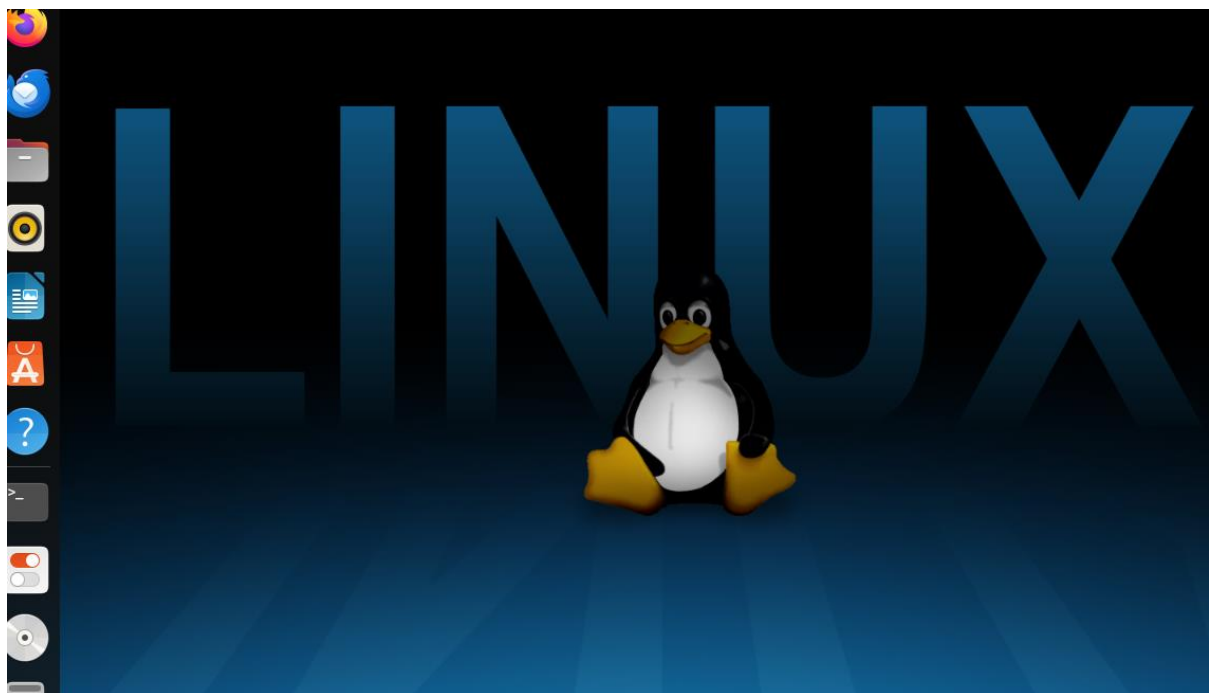
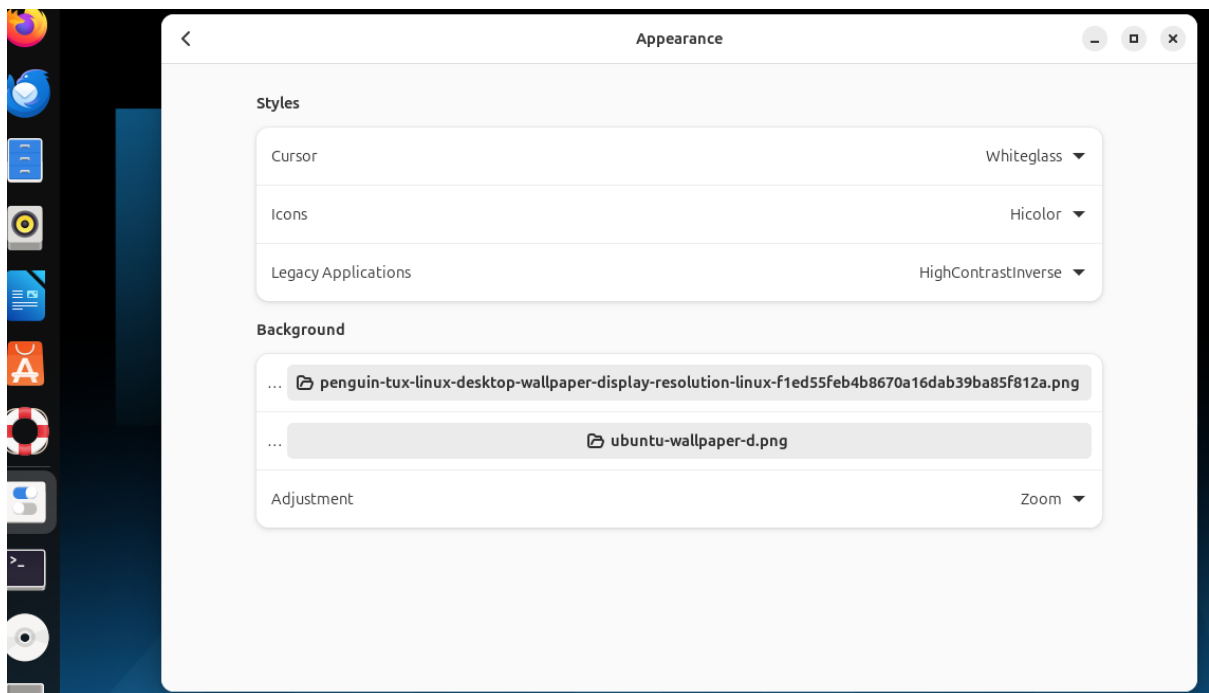
Installing Lock Keys:

- Go to [GNOME Shell Extensions](#)
- Search for Lock Keys in the search bar
- Install the Shell Version of the Lock Keys extension

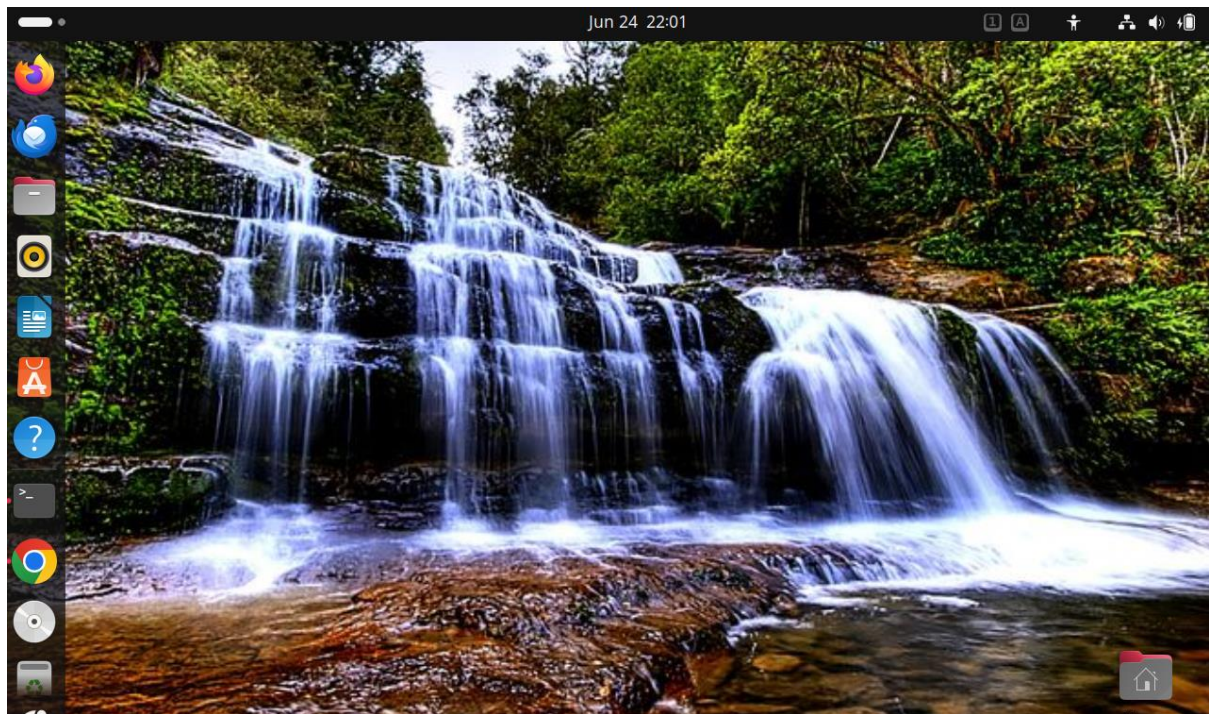
Screenshots of functional Lock Keys:



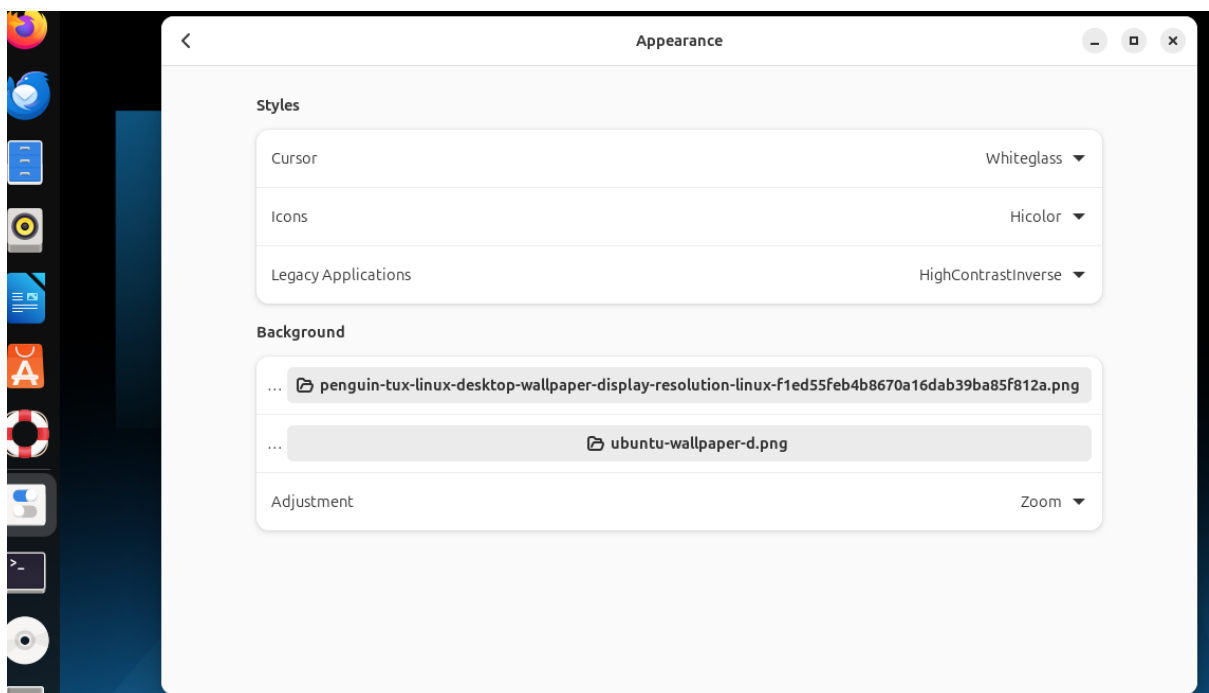
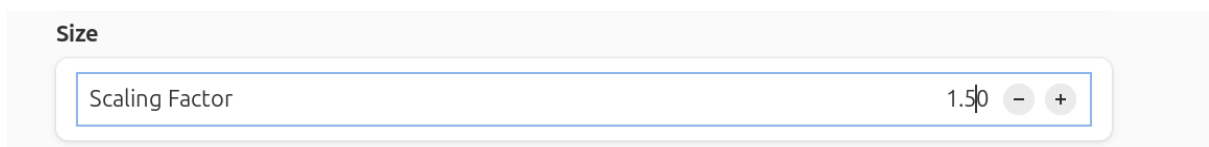
3.4.1



3.4.2



3.4.4



Preferred Fonts

Interface Text	Ubuntu Mono
Document Text	Sans
Monospace Text	Ubuntu Sans Mono

References

Brown, K. (n.d.). *linux config*. Retrieved from linux config web site: <https://linuxconfig.org/how-to-install-gnome-on-ubuntu-22-04-lts-jammy-jellyfish>